Yanghong Liu

stzabl@berkeley.edu | +1 408 505 0455 | GitHub | LinkedIn | Website

EDUCATION

University of California, Berkeley — Expected Graduation: December 2026
B.A. in Statistics, Minor in Data Science — Berkeley, CA

TECHNICAL SKILLS

- **Programming & Tools:** Python, Bash, Git, Linux, R, Transformers, LaTeX
- **Robot Learning & Simulation:** Isaac Sim, Isaac Lab, Reinforcement Learning (RL), PPO, reward design
- **Robotics & Computer Vision:** Video-based 3D reconstruction, object-pose tracking, confidence estimation, teleoperation, camera calibration, policy deployment
- **LLM Training:** LoRA SFT, GRPO, ms-swift, vLLM, DeepSpeed

RESEARCH EXPERIENCE

UC Berkeley — Tomizuka MSC Lab — Berkeley, CA
Research Intern — Robotics & Embodied AI — Dec 2025 – Present
Supervised by Ruihai Wu • UC Berkeley–General Motors research collaboration

PriDex — Learning Bimanual Dexterous Manipulation from Human Videos
Co-first author • Submitted to ICRA 2027 — Jun 2026 – Sep 2026

- Co-developed a pipeline converting egocentric human videos into **physics-validated robot demonstrations** through 3D reconstruction and RL teacher rollouts with task-specific success checks. Across three real-world tasks, our team improved average success from **30%** with real-only training to **63.3%** with mixed real and simulation-generated data.
- Led **confidence-guided residual PPO and reward design**, using 3D-to-2D reprojection and point tracking to estimate reconstruction confidence and adapt exploration. Low-confidence frames allowed larger deviations from reconstructed motion and used human wrist-motion direction rewards to guide learning.

V2AP — Robot Grasp Affordances from Human Videos
First author • Accepted at RSS 2026 RoboData Workshop — Dec 2025 – Jun 2026

- Led **video-to-3D reconstruction and contact extraction** for first- and third-person videos; successful simulated grasps within human-contact regions turned human contact priors into physics-validated grasp labels for robot affordance learning.
- Led **PointNet++ affordance-model training** with posterior supervision; the system used an affordance-conditioned diffusion model to generate 6-DoF grasp poses. Aggregate simulated grasp success improved from **31.9% to 61.6%** over human-prior supervision on unseen object instances and categories.

COURSE PROJECT

Multi-Tool Travel Planning Agent with SFT & GRPO — Mar 2026 – Jun 2026

- Built a **Qwen3-4B-Instruct ReAct agent** with eight tool categories for multi-turn travel planning; augmented data, generated and filtered trajectories, and applied **LoRA SFT followed by full-parameter GRPO**.
- Designed a **six-dimensional curriculum reward** for tool-use compliance, efficiency, and answer quality, using vLLM rollouts and DeepSpeed ZeRO-3 for training. On 80 test tasks, improved average LLM-judged scores by **13.46%** over the Qwen3-4B-Instruct-2507 baseline and **8.53%** over SFT.

INTERNSHIP EXPERIENCE

iDigit — Remote, China
Speech-to-Text & NLP Intern — May 2023 – Aug 2023

- Built and tested a **Whisper-based speech-to-text prototype for users with hearing loss**, integrating audio input, transcription, and grammar and spelling correction into an end-to-end workflow.